Fiscal Year:	FY 2010	**Task Last Updated:**	FY 02/15/2011
PI Name:	Porterfield, D. Marshall Ph.D.		
Project Title:	Early graviperception and polarity in single cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2108	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/01/2009	**End Date:**	04/30/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 4/30/2012 (original end date 4/30/2011), per NSSC (Ed., 4/13/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Roux, Stanley (University of Texas, Austin)		
Grant/Contract No.:	NNX09AH45G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this research will be to understand the fundamental mechanisms associated with the primary process of gravity sensing and the initial stages of signal transduction and cellular polarity development. These studies will focus on the use of Ceratopteris richardii as a model system for gravity sensing. We will focus on the initial mechanoreceptors associated with the trans-cellular calcium current, and will seek to identify the specific membrane component (pumps and channels) associated with these early electrophysiological events using molecular and biochemical approaches. Based on preliminary work we have identified two different annexins and a Ca2+-ATPase as central elements in controlling the gravity-directed flow of calcium through single-celled spores of Ceratopteris. Our project will utilize new technology for advanced-throughput cell electrophysiology, based on novel microfabricated silicon microdevices. We will use this established new method to measure transcellular calcium signaling to further characterize the biophysical properties of gravity sensing and signaling. We are specifically interested in determining how altering the secondary signal transduction signaling events (NO and cGMP) may feedback and control cell responsiveness by altering the primary calcium current. We also will combine the molecular and biophysical approaches in experiments that alter specific gene activities and measure calcium current activities using the lab-on-a-chip systems. The proposed work utilizes a dynamic approach that combines modern molecular cell biology with new advanced-throughput tools for cell physiology, based on silicon microfabricated biosensing. We envision that this will lead to spaceflight experiments that utilize microsatellite opportunities, based on the second generation of the lab-on-a-chip concept. This is based on a centrifugal disk, that will accommodate a variable gravity experiment on orbit. This flight experiment will allow us to directly measure and assay the threshold value for graviperception in the single-cell spores, by directly measuring the transcellular calcium current at variable gravity levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[Editor's note 2/15/2011: No Task Book report received. Progress section and Bibliography compiled from PI's Annual Technical Report.] Gravity directs the orientation and magnitude of a trans-cell calcium current in Ceratopteris spore cells. In order to evaluate what are the likely molecular candidates to serve as calcium channels and pumps that drive this current, we carried out a Next-Generation deep sequencing analysis of all the genes that are expressed in singlecelled Ceratopteris spores after they had been induced to germinate by light, coincident with when the magnitude of the trans-cell calcium current peaks. RNA was extracted from Ceratopteris spores and sent to the sequencing facility at Purdue University (Genomics Core Facility), where it was converted to a cDNA library, which was then sequenced. The sequencing run was successful, and several candidate genes likely to encode CaATPase pumps and potential cation channels were found. In order to determine key molecular events altering cellular signaling a series of experiments will be conducted using the CEL-C system for monitoring transcellular calcium currents in up to 16 cells at a time. There will be several versions of these experiments conducted.
Bibliography Type:	Description: (Last Updated: 03/29/2017)
Abstracts for Journals and Proceedings	ul Haque A, DeCarlo AR, Rokkam M, Roux SJ, Porterfield DM. "Measurements of Gravity Induced Ca2+ Signaling using a Silicon Microfabricated Cell Electrophysiology Lab-on-a-Chip (CEL-C)." 18th Biennial IEEE UGIM (University Government Industry Micro/nano) Symposium, West Lafayette, IN, June 28-July 1, 2010. 18th Biennial IEEE UGIM (University Government Industry Micro/nano) Symposium, West Lafayette, IN, June 28-July 1, 2010. , Jun-2010
Abstracts for Journals and Proceedings	Ul Haque A, Odoemene O, Porterfield DM. "Lab-on-a-chip technologies for Fundamental Biology Research." Physiological Sensing Facility Symposium, West Lafayette, IN, January 2010. Physiological Sensing Facility Symposium, West Lafayette, IN, January 2010. , Jan-2010